

CS Group Project MVC - Code & Presentation Guide

Presentation Reference Document for 5 Modules: Chat, Report, Jasper Report, Notifications, Event Announcements

1. Chat System (Direct & Group Chat with WebSockets)

Overview: Real-time direct & group messaging system using WebSockets for live updates, REST APIs for message operations (send, edit, delete, react, report, block), and database persistence using Hibernate JPA.

File Name	Role / Path	Key Functionality / Notes
ChatController.java	src/main/java/com/hibernate/controller/ChatController.java	REST endpoints for messaging actions (send, edit, delete, react, block, report).
ChatPageController.java	src/main/java/com/hibernate/controller/ChatPageController.java	Page routing for /chat and /chat/room/{id} views.
ChatWebSocketHandler.java	src/main/java/com/hibernate/websocket/ChatWebSocketHandler.java	WebSocket connection session handler & text payload routing.
ChatEventBroadcaster.java	src/main/java/com/hibernate/websocket/ChatEventBroadcaster.java	Broadcasts real-time events (new message, typing, reactions, read receipts).
ChatService.java / Impl	src/main/java/com/hibernate/service/ChatServiceImpl.java	Core business logic for conversation creation, message saving & read status.
MessageMapper.java	src/main/java/com/hibernate/service/MessageMapper.java	Converts Message entities into MessageResponse DTOs.
Conversation.java	src/main/java/com/hibernate/entity/Conversation.java	Entity table storing chat room metadata (direct vs group).
ConversationParticipant.java	src/main/java/com/hibernate/entity/ConversationParticipant.java	Mapping table linking users to chat rooms with roles.
Message.java	src/main/java/com/hibernate/entity/Message.java	Entity storing message text, sender, timestamp, and edit flags.
MessageAttachment.java	src/main/java/com/hibernate/entity/MessageAttachment.java	Stores attached file paths and media metadata.
MessageReaction.java	src/main/java/com/hibernate/entity/MessageReaction.java	Stores user emoji reactions on messages.
MessageSeenStatus.java	src/main/java/com/hibernate/entity/MessageSeenStatus.java	Tracks read/seen status for each room participant.
chat.jsp	src/main/webapp/WEB-INF/views/chat/chat.jsp	Main Inbox UI listing active chat conversations.
room.jsp	src/main/webapp/WEB-INF/views/chat/room.jsp	Active Chat Room UI with WebSocket JS client connection.
create-group.jsp	src/main/webapp/WEB-INF/views/chat/create-group.jsp	UI dialog for starting a new group conversation.

2. Report System (Post, Comment & Message Moderation)

Overview: Multi-tiered reporting system allowing users to report inappropriate posts, comments, or chat messages. Admin dashboard groups reports per item and allows moderation actions (dismiss or delete content).

File Name	Role / Path	Key Functionality / Notes
ReportController.java	src/main/java/com/hibernate/controller/ReportController.java	User API endpoint to submit post reports (/api/reports).
CommentReportController.java	src/main/java/com/hibernate/controller/CommentReportController.java	User API endpoint to report comments (/api/comments/{id}/report).
AdminReportController.java	src/main/java/com/hibernate/controller/AdminReportController.java	Admin Controller to review grouped reports & execute dismiss/delete actions.
ReportService / Impl	src/main/java/com/hibernate/service/ReportServiceImpl.java	Groups multiple reports by target post, calculates counts and reason lists.
AdminCommentService / Impl	src/main/java/com/hibernate/service/AdminCommentServiceImpl.java	Service logic for fetching and managing grouped comment reports.
PostReport.java	src/main/java/com/hibernate/entity/PostReport.java	Entity table storing post report entries.
CommentReport.java	src/main/java/com/hibernate/entity/CommentReport.java	Entity table storing comment report entries.
GroupedPostReportDto.java	src/main/java/com/hibernate/dto/GroupedPostReportDto.java	DTO containing post info, total report count, and list of reporter reasons.
GroupedCommentReportDto.java	src/main/java/com/hibernate/dto/GroupedCommentReportDto.java	DTO containing comment info and aggregated report data.
reports.jsp	src/main/webapp/WEB-INF/views/admin/reports.jsp	Admin Moderation Dashboard UI for reviewing reported posts and comments.

3. Jasper Report Generation (Monthly CheatSheet Report)

Overview: Analytics PDF report generation using JasperReports library. Fetches monthly data, compiles XML template (.jrxml), populates DTO fields, and streams PDF directly to admin browser.

File Name	Role / Path	Key Functionality / Notes
CheatSheetReportController.java	src/main/java/com/hibernate/controller/CheatSheetReportController.java	Admin REST endpoints (/admin/reports/cheatsheet/pdf) to trigger PDF export.
CheatSheetReportService / Impl	src/main/java/com/hibernate/service/CheatSheetReportServiceImpl.java	Compiles JRXML template, fills data source, and converts report to byte[] PDF.
CheatSheetReportDto.java	src/main/java/com/hibernate/dto/CheatSheetReportDto.java	Data Transfer Object holding total posts, total views, active users, top posts.
monthly_cheat_sheet_report.jrxml	src/main/resources/reports/monthly_cheat_sheet_report.jrxml	Jasper Studio XML design template defining tables, headers, styling.

cheatsheet-report.jsp	src/main/webapp/WEB-INF/views/admin/cheatsheet-report.jsp	Admin page UI to select month/year and preview/download PDF reports.
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4. Notification System (User Notifications & Advice)

Overview: In-app notification mechanism triggering alerts for likes, comments, approvals, and announcements. Uses @ControllerAdvice to inject unread counts into all view headers globally.

File Name	Role / Path	Key Functionality / Notes
NotificationController.java	src/main/java/com/hibernate/controller/NotificationController.java	REST endpoints (/api/notifications) for fetching user notifications & mark-read.
NotificationAdvice.java	src/main/java/com/hibernate/controller/NotificationAdvice.java	@ControllerAdvice providing global unread notification count to all JSP views.
NotificationService / Impl	src/main/java/com/hibernate/service/NotificationServiceImpl.java	Core service for building, linking target URLs, and persisting notifications.
Notification.java	src/main/java/com/hibernate/entity/Notification.java	Entity table storing notification message, recipient, sender, type, and read flag.
NotificationRepository.java	src/main/java/com/hibernate/repository/NotificationRepository.java	JPA Repository fetching user notifications sorted by creation date.
notifications.jsp	src/main/webapp/WEB-INF/views/notifications/notifications.jsp	User notification list screen.

5. Event Announcement System

Overview: System-wide announcement management by Admins. Creating an announcement automatically dispatches notifications to all target users and displays banners on user feeds.

File Name	Role / Path	Key Functionality / Notes
AdminAnnouncementController.java	src/main/java/com/hibernate/controller/AdminAnnouncementController.java	Admin REST endpoints (/admin/announcements) for CRUD operations.
AnnouncementService / Impl	src/main/java/com/hibernate/service/AnnouncementServiceImpl.java	Saves announcement and calls NotificationService to alert all users.
Announcement.java	src/main/java/com/hibernate/entity/Announcement.java	Entity table storing title, content, target role, publication status, timestamps.
AnnouncementRepository.java	src/main/java/com/hibernate/repository/AnnouncementRepository.java	JPA Repository for querying active published announcements.
announcements.jsp	src/main/webapp/WEB-INF/views/admin/announcements.jsp	Admin UI for drafting, publishing, and managing announcements.

Presentation & Teacher Defense Key Points

1. Architecture Pattern: The application follows the standard Spring MVC Architecture (Controller Service Repository Database Entity View/JSP). REST endpoints return JSON/PDF streams while page controllers render JSP views.
2. Real-time Communication (WebSocket): Chat functionality uses ChatWebSocketHandler for bidirectional low-latency messaging, combined with ChatService for DB persistence.
3. Grouped Moderation Logic: To prevent duplicate UI items in admin moderation, GroupedPostReportDto aggregates multiple reports targeting the same post into a single row with total report counts and concatenated reasons.
4. Jasper Export Flow: Monthly metrics are compiled into a CheatSheetReportDto, passed to JasperFillManager alongside monthly_cheat_sheet_report.jrxml, and rendered as a PDF byte stream via HTTP Response.
5. Global Notification Advice: NotificationAdvice uses Spring's @ControllerAdvice to inject the unread notification count model attribute into every JSP automatically without repeating code in each controller.